AT A SMOOTH POINT OF A FINITELY PRESENTED ALGEBRA SOME POWER OF AN ELEMENT AVOIDING THE PRIME IS STRICTLY STANDARD

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Editorial note. The source proof is complete; no omitted steps.

Allowed standard prerequisites (editorial).

- Naive cotangent complex of a presented algebra. Algebra, lemma NL-homotopy (cross-chapter, named). Identifies NL_{A/R} with I/I^2 -> direct sum A dx_i for the chosen presentation.
- Localization of the naive cotangent complex. Algebra, lemma localize-NL (cross-chapter, named). Allows evaluating the presenting complex after inverting a.
- Smoothness via a split injection of the conormal module. Algebra, definitions smooth and smooth-at-prime (cross-chapter, named). The split-injection formulation of smoothness at q used to produce a.

Source reference. Original reference 00T2: focused-seeds/geometry/S9-smo-find-strictly-standard/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00TB: focused-seeds/geometry/S9-smo-find-strictly-standard/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00S1: focused-seeds/geometry/S9-smo-find-strictly-standard/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00S7: focused-seeds/geometry/S9-smo-find-strictly-standard/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00S0: focused-seeds/geometry/S9-smo-find-strictly-standard/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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Lemma 0.1. *Let R be a ring. Let $A = R[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$. Let $\mathfrak{q} \subset A$ be a prime ideal. Assume $R \to A$ is smooth at $\mathfrak{q}$. Then there exists an $a \in A$, $a \notin \mathfrak{q}$, an integer c, $0 \leq c \leq \min(n, m)$, subsets $U \subset \{1, \ldots, n\}$, $V \subset \{1, \ldots, m\}$ of cardinality c such that*

$$a = a' \det(\partial f_j / \partial x_i)_{j \in V, i \in U}$$

for some $a' \in A$ and

$$af_\ell \in (f_j, j \in V) + (f_1, \ldots, f_m)^2$$

for all $\ell \in \{1, \ldots, m\}$.

Proof. Set $I = (f_1, \ldots, f_m)$ so that the naive cotangent complex of A over R is homotopy equivalent to $I/I^2 \to \bigoplus A \mathrm{d}x_i$, see Algebra, Lemma 00S1. We will use the formation of the naive cotangent complex commutes with localization, see Algebra, Section 00S0, especially Algebra, Lemma 00S7. By Algebra, Definitions 00T2 and 00TB we see that $(I/I^2)_a \to \bigoplus A_a \mathrm{d}x_i$ is a split injection for some $a \in A$, $a \notin \mathfrak{q}$. After renumbering $x_1, \ldots, x_n$ and $f_1, \ldots, f_m$ we may assume that $f_1, \ldots, f_c$ form a basis for the vector space $I/I^2 \otimes_A \kappa(\mathfrak{q})$ and that $\mathrm{d}x_{c+1}, \ldots, \mathrm{d}x_n$ map to a basis of $\Omega_{A/R} \otimes_A \kappa(\mathfrak{q})$. Hence after replacing a by aa' for some $a' \in A$, $a' \notin \mathfrak{q}$ we may assume $f_1, \ldots, f_c$ form a basis for $(I/I^2)_a$ and that $\mathrm{d}x_{c+1}, \ldots, \mathrm{d}x_n$ map to a basis of $(\Omega_{A/R})_a$. In this situation a^N for some large integer N satisfies the conditions of the lemma (with $U = V = \{1, \ldots, c\}$). $\square$